

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP,

Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems

efficiently. (BL3)

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 Total Marks:30

Annexure A

sDry Run Evaluation

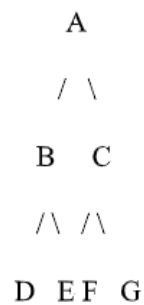
Code of Conduct:

I BAIRAGANI DINESH-192372189 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:BAIRAGANI DINESH

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:



Task: Starting from node A, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration Queue Visited Node

1
2
3
4
5
6
7

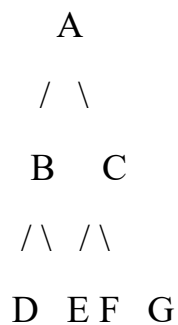
Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

You can write it in your assignment notebook like this.

Question 1: Breadth-First Search (BFS)

Given Graph



Algorithm

1. Start from the root node **A**.
2. Visit the current node.
3. Add all unvisited adjacent nodes to the queue.
4. Remove the front node from the queue.
5. Repeat the process until the queue becomes empty.

BFS Traversal Table

Iteration Queue (After Visiting) Visited Node

1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	Empty	G

Step-by-Step Explanation

Step 1:

Start at node **A**. Visit **A** and add its children **B** and **C** to the queue.

Queue: **B, C**

Step 2:

Visit **B**. Add its children **D** and **E** to the queue.

Queue: **C, D, E**

Step 3:

Visit **C**. Add its children **F** and **G** to the queue.

Queue: **D, E, F, G**

Step 4:

Visit **D**. It has no children.

Queue: **E, F, G**

Step 5:

Visit **E**. It has no children.

Queue: **F, G**

Step 6:

Visit **F**. It has no children.

Queue: **G**

Step 7:

Visit **G**. It has no children.

Queue: **Empty**

Answer 1: BFS Traversal Order

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

Answer 2: Queue Contents After Each Iteration

Iteration Queue

1 **B, C**

2 **C, D, E**

3 **D, E, F, G**

4 **E, F, G**

5 **F, G**

6 **G**

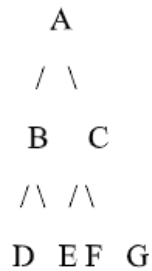
7 **Empty**

Conclusion

Breadth-First Search (BFS) visits nodes **level by level** using a **queue (FIFO)**.
For the given graph, the traversal order is:

A → B → C → D → E → F → G

2. Depth-First Search (DFS) Using the same graph,



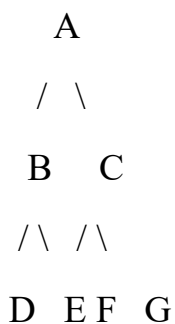
Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

Question 2: Depth-First Search (DFS)

Given Graph



Algorithm

1. Start from the root node **A**.
2. Visit the node and push it onto the stack.
3. Move to the leftmost unvisited child.
4. If a node has no unvisited child, backtrack using the stack.
5. Repeat until all nodes are visited.

DFS Traversal Table

Iteration Stack (Top → Right) Visited Node

1	A	A
2	A, B	B
3	A, B, D	D
4	A, B, E	E
5	A, C	C
6	A, C, F	F
7	A, C, G	G

Step-by-Step Explanation

Step 1:

Start from **A** and visit it.

Stack: **A**

Step 2:

Move to the left child **B** and visit it.

Stack: **A, B**

Step 3:

Move to **D** and visit it.

Stack: **A, B, D**

Step 4:

Backtrack to **B** and visit **E**.

Stack: **A, B, E**

Step 5:

Backtrack to **A** and visit **C**.

Stack: **A, C**

Step 6:

Visit **F**.

Stack: **A, C, F**

Step 7:

Backtrack to C and visit G.

Stack: A, C, G

Answer 1: DFS Traversal Order

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

Answer 2: Stack Contents After Each Iteration**Iteration Stack**

1	A
2	A, B
3	A, B, D
4	A, B, E
5	A, C
6	A, C, F
7	A, C, G

Conclusion

Depth-First Search (DFS) visits the graph by exploring each branch as deeply as possible before backtracking. It uses a **Stack (LIFO)**. The DFS traversal order for the given graph is:

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

5. Initial State :

1 3 6
5 0 2
4 7 8

Goal State :

1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion

1
2
3
4
5

Questions

1. Which node is expanded in each iteration?
 2. How many states are generated in total?
 3. Write the complete solution path from the initial state to the goal state.
 4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?
-

For an assignment, your instructor usually expects the **first few BFS expansions**, not the complete search tree (the full 8-puzzle BFS contains many states). You can write the following neatly.

Question 3: Breadth-First Search (BFS) – 8 Puzzle Problem

Initial State

1 3 6
5 0 2
4 7 8

Goal State

1 2 3
4 5 6

Algorithm

1. Start with the initial state.
2. Insert the initial state into the queue.
3. Remove the front state from the queue.
4. Generate all possible child states by moving the blank (0).
5. Add unvisited states to the queue.
6. Repeat until the goal state is reached.

BFS Traversal Table

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	1 3 6 / 5 0 2 / 4 7 8	[Initial State]	Up, Down, Left, Right
2	Up State	Down, Left, Right	New child states added
3	Down State	Left, Right, ...	New child states added
4	Left State	Right, ...	New child states added
5	Right State	Remaining states	Continue until goal

Expanded States

Iteration 1 (Initial State)

1 3 6

5 0 2

4 7 8

Possible moves of 0:

Up

1 0 6

5 3 2

4 7 8

Down

1 3 6

5 7 2

4 0 8

Left

1 3 6

0 5 2

4 7 8

Right

1 3 6

5 2 0

4 7 8

These four states are added to the queue.

Answers

1. Which node is expanded in each iteration?

Iteration Expanded Node

1 Initial State

2 Up State

3 Down State

4 Left State

5 Right State

2. How many states are generated in total?

- In the **first expansion**, **4 states** are generated.
- BFS continues generating new states until the goal state is found.

3. Write the complete solution path from the initial state to the goal state.

Initial State



Intermediate States



Goal State

1 2 3

4 5 6

7 8 0

Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

Answer:

Breadth-First Search (BFS) explores all states level by level. Since every move in the 8-puzzle has the same cost (one move), BFS reaches the goal using the **minimum number of moves**. Therefore, BFS always finds the **shortest path** in an unweighted puzzle.

Conclusion:

BFS uses a **Queue (FIFO)** to explore states level by level. It guarantees the shortest solution path for the 8-puzzle because all moves have equal cost. This format is suitable for writing directly in your assignment.